Indoor Localization for Car Finding

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ABSTRACT

Indoor positioning provides a new experience for the users, and becomes more and more important for the car-finding in the underground garages. Researchers have come up with many accurate indoor localization systems in the laboratory. However, they can hardly been deployed in the garages due to many challenges in real world. In this work, we propose a smartphone-based indoor positioning system **LocateMe**, which leverages radio signals and geomagnetic fields to provide fingerprinting-based localization service in garages. In case of the poor network service, we calibrate the dead reckoning by using the Bluetooth Low Energy signals. Our experiments show that the median accuracy of **LocateMe** can reach 5 meters in both controlled and in-the-wild scenarios.

1 INTRODUCTION

When using the positioning service, most people reach their destination by using turn-by-turn navigation. Almost all the navigation systems used in outdoor environments can be provided by GPS or other GNSS[16]. However, as these systems require satellite signal reception, their performance is hugely declined in indoor environments due to the occlusion of the complex building structure.

Meanwhile, Indoor positioning is an emerging topic and obtains many concentrations from smartphone users and business owners. For users, indoor localization provides a better user experience in a large building (*e.g.,* garage). For business owners, improving user experience by indoor localization is critical to attracting more customers. In this work, we focus on the underground car-finding system, an emerging indoor localization application. Unlike the positioning during shopping, which mall owners might contradict since they prefer users to spend more time in the mall, the car-finding service is welcomed by both users and mall owners because users can waste less time outside the mall.

From the perspective of data sources, indoor positioning technologies are mainly divided into three categories. The first category is Wireless-based positioning, such as Wireless Local Area Network (WLAN) methods, ZigBee-based methods [8, 12], Bluetooth-based methods [5, 7, 33], Ultra Wide Band (UWB) methods [1, 2]. The second category is Measuring sensor-based positioning, such as Pedestrian dead reckoning (PDR), Geo-magnetic field [9, 27], Radio Frequency (RF) proximity sensors [3, 21, 26]. The third category is Vision-based positioning, like visible light methods [14, 28]. The above techniques have some shortcomings:

- Cellular data is usually limited due to poor signal strength, and WiFi service is often unavailable due to the lack of infrastructure. We cannot use techniques that consume too much data (*e.g.,* computer vision).
- The signal fading model inside the building is complex due to non-line-of-sight landmarks, multipath signal transmission, and signal noises.
- Some state-of-the-art works leverage the channel state information (CSI) of radio signals (*e.g.,* WiFi) to achieve direction finding, but they need the support of dedicated hardware.
- The cost of such a system is a big concern for deployment. We need to leverage lightweight hardware to achieve accurate indoor localization.

To overcome the defects of the single method and make the indoor localization system robust in extreme environments, we propose a practical car-finding system **LocateMe** based on Bluetooth low energy (BLE) beacons and geomagnetism in this paper. It first achieves coarse localization with BLE beacons, and further combine geomagnetic fields (GMF) with BLE to provide fine-grained positioning service based on particle filtering. In this work, we aim at finding the cars rather than the parking space by locating the user. Note that the position of the car is given by the other project (not discussed in this project) which leverages camera to find the position of the car.

We conduct several experiments to validate the fingerprint source we choose, use dynamic time warping(DTW) [6] to achieve deal with the device heterogeneity, and further leverage BLE beacons to calibrate the result of dead reckoning (DR). Considering that there may or may not be cars in the parking spaces, we first test the effect of cars on the GMF. Our experiments show that the magnetic fields around the car remain stable even if we are around 0.5m from the car. When matching the fingerprints, we choose to compare the trend of features rather than the Euclidean distances since the readings on signal strength differ among devices. Further, we leverage the trend of signal strength to estimate the position of the user and calibrate the errors in DR, which is essential when the network is unavailable. Finally we conduct many experiments to evaluate our system in both controlled and real-world scenarios.

The rest of the paper is organized as follows. Challenge explains basic concepts and related works, and introduces the problem we aim to solve in this project. Design details the methodology adopted for the robust positioning service of the car-finding system in underground parking garages. The experimental results and discussion are in Evaluation, followed by the individual contribution, and conclusions.

2 CHALLENGE

In this paper, we make use of fingerprinting to provide the indoor positioning service. To achieve this goal, there needs to be a feature map and the matching system. The feature map is built to store the collected signal data, including radio signals and geomagnetic fields. The matching system needs to deal with data errors from multiple sources. Note that we only focus on the matching system rather than the building of feature maps.

The positioning is based on the matching between scanned data and the feature map. In short, our system should find the position which has most similar features with the measured ones by the smartphone. The problem is that the signals usually change with time, and the measurement of the same signal at the same position may differs among smartphones [29]. As shown in Fig. 1, indoor signals not only vary along with the distance, but can be affected by shadow and multi-path scenarios. These phenomenons make the classical signal fading model [25] not suitable for the distance estimation in the underground garages.

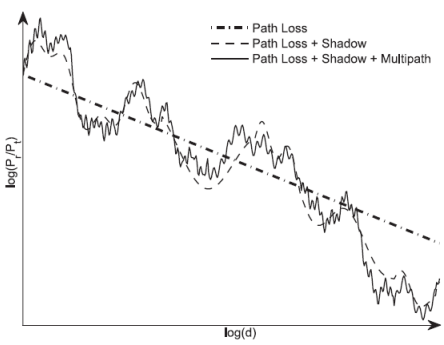


Figure 1: Signal fading with respect to distance (Fig. 1 of [29]). In real-world scenarios, signal strength will be affected by not only distance but also noises.

The data scanning by different smartphones will also have different results due to the difference of their hardware and software. For example, the measurement of the geomagnetic field at the same location differs between smartphones (Fig. 2). In fact, it is impossible for us to collect the data by all the off-the-shelf smartphones. Since we rely on smartphones to scan signal features in the mall, measurement deviations will significantly affect the localization performance. In other words, our fingerprinting-based system cannot directly use the measured features (*e.g.,* geomagnetic fields) for indoor positioning.

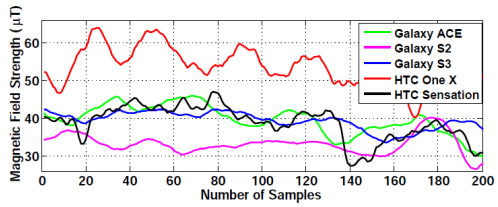
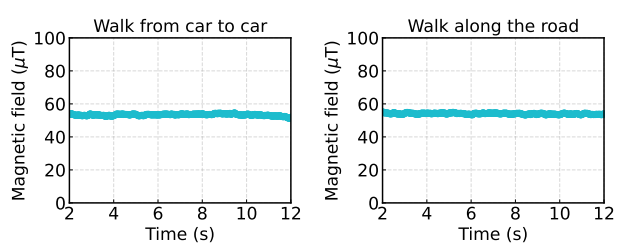


Figure 2: Deviation of signal measurement by different smartphones (Fig. 14 of [31]).

Another concern about the GMF features is their stability. There may or may not be a car in the parking space, which will affect the geomagnetic field. In other words, the GMF features in the garage might be unstable. Considering that, we set up an experiment to evaluate the effect of cars on GMF. We first walk across the road from a car to another car, and then walk along the road. During walking, we use the smartphone to measure the geomagnetic fields. The experimental results demonstrate that the cars have no obvious effect on the geomagnetic fields. As shown in Fig. 3a and Fig. 3b, the magnetic field is always around 53 μT no matter if we walk across the road from car to car, or walk along the road. In fact, our further tests show that only when the distance between the smartphone and the car is within 20cm, the car will affect the GMF measured by the smartphones. In real-world cases, the user will not get too close (*e.g.,* less than 50cm) to the car when walking on the road. In other words, the GMF in the garage can be used as a stable fingerprinting feature.

In summary, we need to propose a practical indoor positioning system to locate the users during finding their cars in the underground garages. It will deal with the complex



(a) Across the road **(b) Along the road**

Figure 3: The affect of the cars on geomagnetic fields.

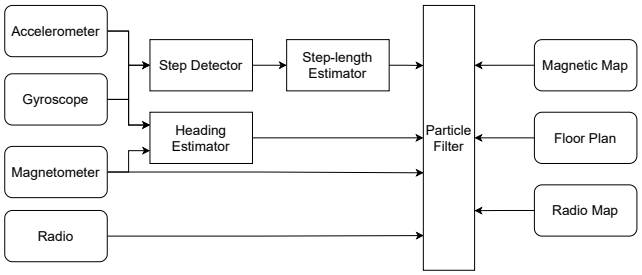


Figure 4: Algorithm framework

signal fading model, device heterogeneity, and the noises in radio signals, geomagnetic fields and inertial data.

3 DESIGN

The basic idea of our algorithm is to match the measurement against the signal map (*i.e.,* fingerprinting). As shown in Fig. 4, our positioning system is based on particle filtering [10]. We first calculate the coarse position of the user, and then leverage particle filtering to obtain the fine-grained position and thus track our user. During particle filtering, **LocateMe** utilizes inertial sensors to calculate the steps of the user, and combines magnetic data to further obtain the heading. Taking into account the differences among devices, we use dynamic time warping (DTW [6]) to make fingerprint matching rather than just compare the Euclidean distance. Furthermore, we leverage BLE signals to improve the performance of dead reckoning, which is essential to the positioning service when network is unavailable.

3.1 Dead reckoning

In dead reckoning, we focus on the detection of step detection and estimation of step length. Before detecting step, we use the 4th order Butterworth low-pass filter to remove the noise in the measurement results. As shown in Fig. 5, the signal obtained is a periodic signal that is almost sine wave, and the periodicity of the acceleration signal in the walking state can be clearly observed. Pedestrians take two steps per second by default, so we set the cutoff frequency of the filter to 2. The other needs to be subtracted from the sine wave is the value of gravitational acceleration. Now we get a waveform that can directly reflect the walking state of a user.

Step detection is to find out the peak and valley of the acceleration curves. According to our experimental results, the peak threshold is 2.0 and the trough threshold is 0.6. Finally, we obtain a list of discrete values (1, 0, and -1), and utilize the values in a certain window (sample size of a step) to find the step. The signal less than 0.6 is regarded as -1, the signal greater than 2.0 is regarded as 1, and the middle signal between them is regarded as 0. Observing the

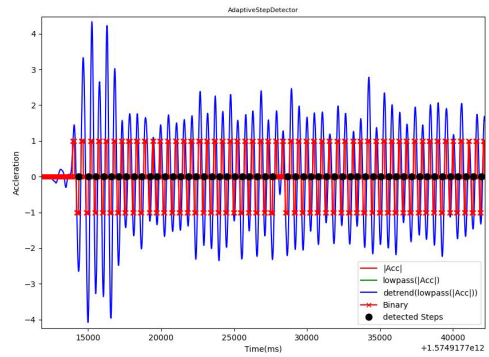


Figure 5: Step detection. When the users are walking, their acceleration will change periodically. We leverage the change of the acceleration to detect the steps.

signal, we can know that there are only one peak and one trough in a walking cycle. The acceleration change from 1 to -1 is considered one step. In this way, we obtain the step frequency of the user.

The prediction of the step length uses the following empirical formula. acc_{max} is the maximum of the acceleration, and acc_{min} is the minimum of the acceleration. The key here is the parameter k. According to our measurement, k should be set to $0.4 \sim 0.45$.

$$l = \sqrt[4]{(acc_{max} - acc_{min})} * k \tag{1}$$

Step length l will be used in our positioning algorithms.

In fact, the model of step-length estimation in Eq. (1) is not accurate in real-world scenarios. Therefore k will be adjusted during positioning. Our method is to leverage the BLE beacons to further calibrate our model, which will be introduced in Section 3.3.

3.2 Localization

The coarse position of users is obtained by using trilateration. Given the distance of the user to three different beacons, we can calculate the coordinates of the user. Based on the signal fading model, when we have collected three different signals, we can use them to calculate the distance to the corresponding beacons (whose positions are already known), and further use trilateration to compute the user's coarse position. When the user moves, **LocateMe** will leverage particle filtering [10] to calculate the location of the user.

The pedestrian movement is regarded as the state transition in particle filtering, and the measurements in the observation equation refer to the fingerprints, including GMF and BLE features. The weight for each particle depends on the similarity between the currently measured fingerprint and the features at its location in the signal map. The similarity

is inversely proportional to the vectorized distance calculated based on the dynamic time warping (DTW) algorithm [6].

We have incorporated the step number with DTW algorithm to improve its performance (Algorithm 1). DTW can measure the similarity of two temporal sequences and is usually used in speech recognition. **LocateMe** leverages DTW to match two point sequences in its particle filtering, which is much more reliable than just comparing two points. N is the sequence length, *i.e.,* the number of steps. When the errors in dead reckoning (or inertial navigation) is negligible, a longer point sequence helps improve the performance of DTW and particle filtering. As the step number N increases, errors will accumulate. When the deviation δ in particle filtering is beyond limit, we reset N to be N_{min}. In conclusion, we increase the step number N by N_{delta} when the positioning deviation δ is small and reduce N when δ is beyond limit. In our implementation, N_min, N_{delta}, and N_{max} are 4, 4, and 12 respectively. The errors of inertial navigation in a short distance are acceptable, hence particle filtering is not needed for each step. **LocateMe** runs the filtering (including dynamic-step DTW) every 4 steps.

Algorithm 1: Fingerprint matching.

```
// δ is the deviation in particle filtering;
Input: test point p, user track tk, δ
Result: Similarity sim between p and tk[−1]
// N is the number of steps for matching;
if δ ≥ limit then
    N ← N_min;
else
    N ← max(N + N_delta, N_max);
end
// use the last N points for testing;
t ← tk[−N :];
// retrieve the points in the map as reference r;
// r ends with point p and its track is the same to t;
r ← retrieve(t, p);
// match their similarity between the feature
  sequences of t and r;
sim ← DTW(t, r);
return sim;
```

3.3 Dead Reckoning Calibration

When the network service is unavailable, we have to rely on dead reckoning to achieve indoor positioning. Note that the network is oftentimes not good enough in the underground garage. Dead reckoning has accumulated errors, which means we walk longer, the positioning errors become larger. We need to improve the performance of dead reckoning. When the users use **LocateMe**, however, there is no ground truth about their position, making it difficult to calibrate the result of dead reckoning.

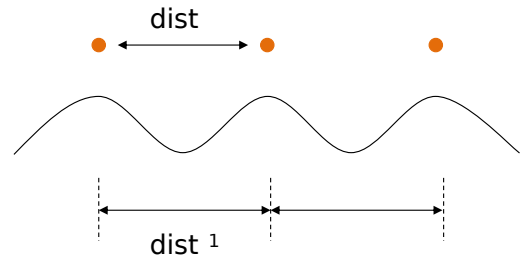


Figure 6: The RSS trend of beacons. The red circles represent the beacons (placed on the ceiling), and the wave represents the received signal strength (RSS). Please note that when the user is directly below the beacon, the RSS of the beacon reaches its maximum value. So $dist^1$ is approximately equal to $dist$.

Since there are beacons deployed on the ceiling directly above the road, we choose to leverage these beacons to calibrate the dead reckoning. According to the fading model of the radio signal, the closer you are to the beacon, the stronger the RSS will become. As shown in Fig. 6, the red circles represent the beacons (placed on the ceiling), and the wave represents the received signal strength (RSS). When the user is directly below the beacon, the RSS of the beacon reaches its maximum value. In this case, the distance $dist^1$ between the positions where RSS reaches the maximum value is approximately equal to the distance $dist$ between the positions of two corresponding beacons.

In fact, due to the limitation of the scanning frequency of smartphones, we oftentimes cannot directly measure the maximal RSS of the beacon. We use the curve-fitting to recover the whole curve of RSS with respect to distance based on the signal fading model. After that, We can obtain a very approximate value ($dist$) of the user's walking distance ($dist^1$). Note that we use Eq. (1) to estimate the step length of the user, therefore we can use $dist$ to calibrate the parameter k. For example, our dead reckoning based on k_{old} shows the walking distance is A, while the ground truth is around A'. Thus we can update $k_{new} = k_{old} * \frac{A'}{A}$.

4 EVALUATION

We evaluate our system in both controlled and in-the-wild scenarios. The controlled experiments are conducted in a company office which is $15m \times 40m$. Beacons are deployed on the office floor every 4 meters, and the broadcast interval is 200ms. We further test **LocateMe** in a real-world garage,

which is $200m \times 300m$. Beacons are deployed on the ceiling above the garage road every 10 meters, and the broadcast interval is 200ms. Besides, our experiments are conducted using OPPO (Android 8.0) if not specified.

4.1 Dead Reckoning

We evaluate the performance of our estimation on step and its length based on the crowdsourcing data that is used to construct the feature map of signals (e.g. geomagnetic field) in the building. Step detection and step-length estimation are essential to the dead reckoning, which is leveraged in **LocateMe** to enhance the positioning accuracy. The collector marks the location of himself during scanning the signals to assign them to the corresponding building position at intervals. Besides, we together with the mall staff will measure the data and test the performance of our system before delivering the project. In this way, we have obtained the ground truth about the position in which the signals are measured.

The estimation on step length is adjusted when the user walks. At the beginning, we do not have the knowledge about the step characteristics of the user, thus there are sometimes large biases in these steps. As shown in Fig. 7, the bias in the first two steps is much larger than that in the next steps. The next estimation on the step length becomes more accurate, and the estimated step length is around 0.8 $\sim$ 1.25 of the ground truth.

It is in fact hard to compute the accurate length of each step, therefore we update the observation results every certain steps (usually four), rather than one step, when using particle filtering. Another concern is about the heading errors. In the scenario of underground garages, the walking area is usually limited and we can leverage the knowledge of map to remedy the user direction. We also find that our system might have large errors in estimating the direction when the user pauses and changes his heading in the intersection. Due to errors in the compass and gyroscope data, the pause and wag will lead to the incorrect calculation of user direction. Hence it is necessary to test all the possible paths when the user pauses and swings at the crossing.

4.2 Positioning accuracy

We evaluate the positioning performance of **LocateMe** by using the crowdsourcing traces and our measurement data. The collectors mark their location at intervals in crowdsourcing and our measurement, thus there is ground truth for these data.

LocateMe leverages BLE and geomagnetic fields (GMF) to locate the user (or smartphone), and the results in the office are shown in Fig. 8. The bias of BLE-based positioning (Fig. 8) is nearly 1 $\sim$ 2 meters larger than that of the BLE and

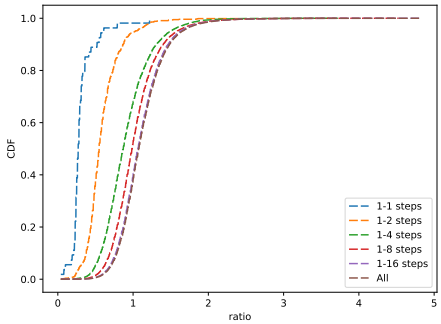


Figure 7: The errors in the estimation of step length.

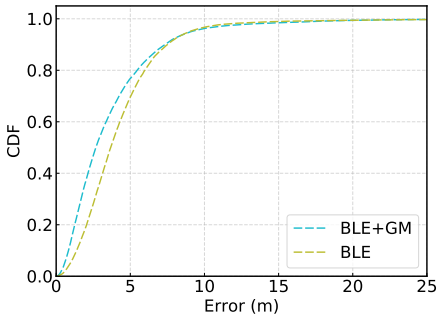


Figure 8: Positioning accuracy in the office (controlled experiments).

GMF-based algorithm. In most cases, geomagnetic fields can improve the accuracy of positioning since it provides addition fingerprinting feature. But the improvement brought by GMF is not limited (1 $\sim$ 2 meters), because the resolution of GMF is limited due to the heterogeneity of smartphones and their sensitivity.

In the laboratory, the resolution of GMF features collected by professional equipment can reach the centimeter level, which means the positioning accuracy can also reach the centimeter level. However, the feature resolution of GMF collected by smartphones can only reach the meter level (around 2 meters). There are also long tail errors, which are caused by the lacking of BLE signals and large geomagnetic resolution. In extreme cases, the performance of BLE and GMF-based algorithm is even worse than BLE-based algorithm. GMF features are not globally unique in the floor, and thus will confuse the positioning result provided by BLE features if the position errors are too large. Besides, geomagnetic fields are similar between symmetric areas (*e.g.,* fork in the road). If the smartphone does not collect strong signals to correct the positioning results, the accumulated error will continue to increase over time.

In real world garages, as illustrated in Fig. 9, the median positioning errors can be less than 5 meters. Compared to

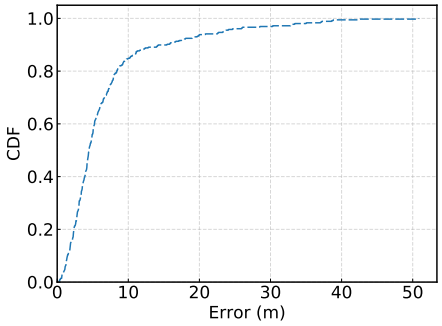


Figure 9: Positioning accuracy in the garage (real-world experiments).

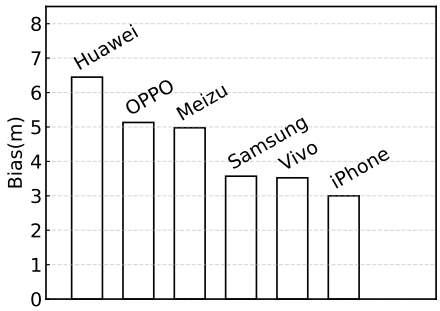


Figure 10: Positioning accuracy of different devices.

the experiment in the office, the distance between beacons in the garage is farther, leading to less strong signals scanned by the smartphones. Therefore, the positioning errors also become larger. But the accuracy of 5 meters is still enough for users to find their cars. The long tails in Fig. 9 are mainly caused by the absence of BLE signals (or strong signals). We cannot obtain the RSS of the specific beacons every time we scan the BLE signals. For example, We may only measure the RSS of each beacon twice per minute (note that we get a lot of RSS of many different beacons). In this case, the useful RSS (*i.e.,* strong RSS) might be not enough for localizing our target. We further test the performance of our service by using different devices, and the results are shown in Fig. 10. The positioning accuracy of iPhone is about 3.0 meters, which is the best among all devices. Huawei performs worst among the Android devices, which is caused by its scanning ability. To save energy, Huawei smartphones choose to limit their scanning frequency of radio signals (both WiFi and Bluetooth), and this will reduce the useful features collected by the smartphones during scanning. Besides, the inertial data of some devices (*e.g.,* Meizu) is not accurate due to the limit of their hardware.

5 RELATED WORK

To provide navigation service for the users, various methods have been proposed by the researchers.

BLE Beacons: Bluetooth low energy (BLE) beacons are small transmitters that periodically send unique data packets. Received signal strength indicator (RSSI) can be used to determine the location of the beacons [24]. Compared to alternative technologies such as WiFi, beacons are cheap, small and easy to place. Filtering algorithms are used to improve the accuracy of RSSI-based techniques [20]. Zafari *et al.*. [30] apply server-side Kalman Filter to improve BLE proximity estimation. In an indoor beacon-based positioning system, Zhang *et al.* [32] test three filtering techniques: Kalman filter, Gaussian filter, and Gauss-Kalman filter. The Gauss-Kalman filter has the best performance. Mackey *et al.* implement BLE and particle filtering in a small parking system [19].

Geomagnetism: Each position has its own characteristics of the magnetic field. Geomagnetism identifies the characteristics of magnetic field signatures as reference fingerprints to locations, and matches the real-time magnetic field to the reference fingerprint to find the corresponding position [9]. The geomagnetic matching algorithm is the core technology of Geomagnetism. There are many matching algorithms, such as MSD, MAD, and PORD [17]. We utilize dynamic time warping(DTW) for geomagnetism matching. Several papers have implemented this method. Qiu *et al.* leverage particle filtering and DTW to track the target in the path matching mode [23]. Lee *et al.* implement DWT in their MTS-enhanced system to improve accuracy [15].

Dead Reckoning: The Dead reckoning (DR) algorithm detects the occurrence of steps and calculates the distance traveled. With direction of movement and distance traveled, trajectory points can be generated. In mobile devices, like phone, gyroscopes and magnetometers are popularly used to estimate the heading, and accelerometer and gyroscopes are used popularly used to estimate step [11]. Dead reckoning has been widely used in indoor positioning system. In 2007, Ojeda *et al.* implement personal Dead reckoning system for GPS-denied environments [22].

Hybrid System: Since there are multiple indoor positioning techniques and each technique has its own defect, the popular method now is combining these methods according to the needs of the application. Lu *et al.* [18] design a hybrid indoor positioning algorithm based on WiFi fingerprinting and pedestrian Dead reckoning. Kuang *et al.* [13] propose an indoor positioning system based on pedestrian Dead reckoning and magnetic field matching. Ban *et al.* [4] propose a high accuracy indoor positioning method that uses residual magnetism, pedestrian Dead reckoning and WiFi-based localization methods. However, all these systems lack

comprehensiveness. Compared to **LocateMe**, some lack the coarse localization that can speed up the localization, and others do not adopt the essential GMF matching algorithm that fits for different mobile devices.

6 INDIVIDUAL CONTRIBUTION

In this work, Qi Le is responsible for conducting the survey about the background and related works. Qi further set up the experiments to test the effect of cars on the geomagnetic fields. Yuming Hu implements the dynamic-step fingerprinting algorithm based on common modules, and leverages BLE signals to calibrate the models of dead reckoning. In addition, Qi and Yuming together analyze the experiment results. As for the project report, Qi is responsible for the Introduction and Related-work section, and Yuming is in charge of the Challenge, Design and Evaluation section.

7 CONCLUSION

We propose a fingerprinting-based indoor positioning system **LocateMe** that is practical for underground garages. It leverages geomagnetic fields and Bluetooth low energy to estimate the location of smartphone users. We further leverage beacons to calibrate the dead reckoning model so that our system can still operate normally when network services are unavailable. **LocateMe** can provide robust localization service at low cost since it only needs the BLE beacons to be deployed in the garages and the experiments show that its median error can be less than $5m$ in the real-world garage.

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